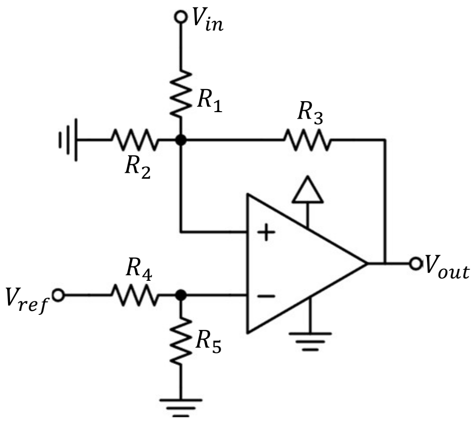
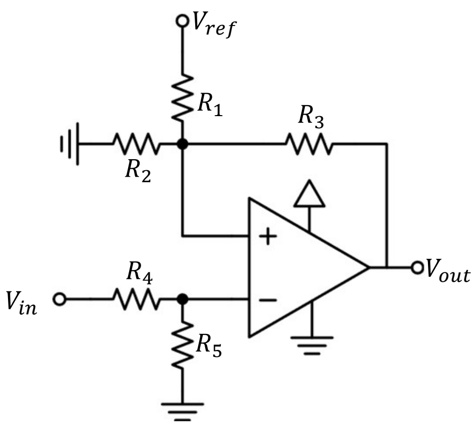
Non-Inverting Comparator



Inverting Comparator



Solution Non-inverting

$$y = 1, x < 1$$

$$x = \frac{V_{ref}V_{DD}}{V_{inH}V_{DD} - V_{ref}V_{inH} + V_{ref}V_{inL}}$$

$$z = \frac{y}{x}\frac{V_{ref}}{V_{inH}}$$

- R_3: choice

$$R_1 = \frac{1-z}{xz}R_3$$

$$R_2 = \frac{x}{1-x}R_1$$

Normalise, tolerance, etc

$$\boxed{x = \frac{R_2}{R_1 + R_2}}$$

$$\boxed{z = \frac{R_3}{xR_1 + R_3}}$$

$$V_{inH} = \frac{yV_{ref}}{xz}$$

$$V_{inL} = \frac{yV_{ref} - (1-z)V_{DD}}{xz}$$

Solution Non-inverting

$$x = 1, y < 1$$

$$y = \frac{V_{inH}V_{DD}}{V_{ref}V_{DD} + V_{ref}V_{inH} - V_{ref}V_{inL}}$$

$$z = \frac{y}{x}\frac{V_{ref}}{V_{inH}}$$

- R_3: choice

$$R_1 = \frac{1-z}{z}R_3$$

- R_4: choice

$$\boxed{R_5 = \frac{y}{1-y}R_4}$$

Normalise, tolerance, etc

$$\boxed{y = \frac{R_5}{R_4 + R_5}}$$

$$\boxed{z = \frac{R_3}{xR_1 + R_3}}$$

$$V_{inH} = \frac{yV_{ref}}{xz}$$

$$V_{inL} = \frac{yV_{ref} - (1-z)V_{DD}}{xz}$$

Solution Inverting

$$y = 1, x < 1$$

$$x = \frac{V_{inL}V_{DD}}{V_{inL}V_{ref} + V_{ref}V_{DD} - V_{in}\ \ V_{ref}}$$

$$z = \frac{y}{x}\frac{V_{inL}}{V_{ref}}$$

- R_3: choice

$$R_1 = \frac{1-z}{xz}R_3$$

$$R_2 = \frac{x}{1-x}R_1$$

Normalise, tolerance, etc

$$\boxed{x = \frac{R_2}{R_1 + R_2}}$$

$$\boxed{z = \frac{R_3}{xR_1 + R_3}}$$

$$V_{inL} = \frac{xz}{y}V_{ref}$$

$$V_{in} = \frac{xzV_{ref} + (1-z)V_{DD}}{y}$$

Solution Inverting

$$x = 1, y < 1$$

$$y = \frac{V_{ref}V_{DD}}{V_{inH}V_{ref} + V_{inL}V_{DD} - V_{inL}V_{ref}}$$

$$z = \frac{y}{x}\frac{V_{inL}}{V_{ref}}$$

- R_3: choice

$$R_1 = \frac{1-z}{z}R_3$$

- R_4: choice

$$\boxed{R_5 = \frac{y}{1-y}R_4}$$

Normalise, tolerance, etc

$$\boxed{y = \frac{R_5}{R_4 + R_5}}$$

$$\boxed{z = \frac{R_3}{xR_1 + R_3}}$$

$$V_{inL} = \frac{xz}{y}V_{ref}$$

$$V_{inH} = \frac{xzV_{ref} + (1-z)V_{DD}}{y}$$